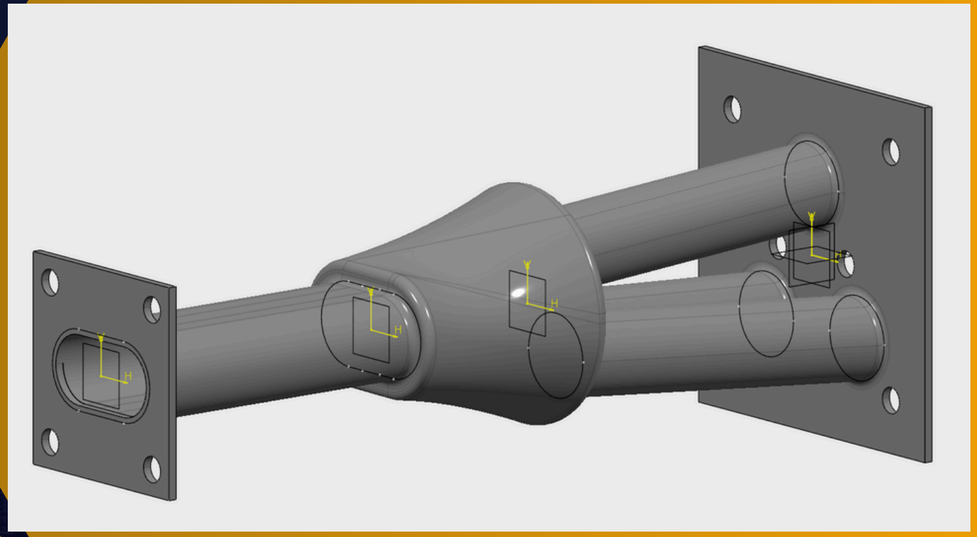
Design of a hydraulic manifold

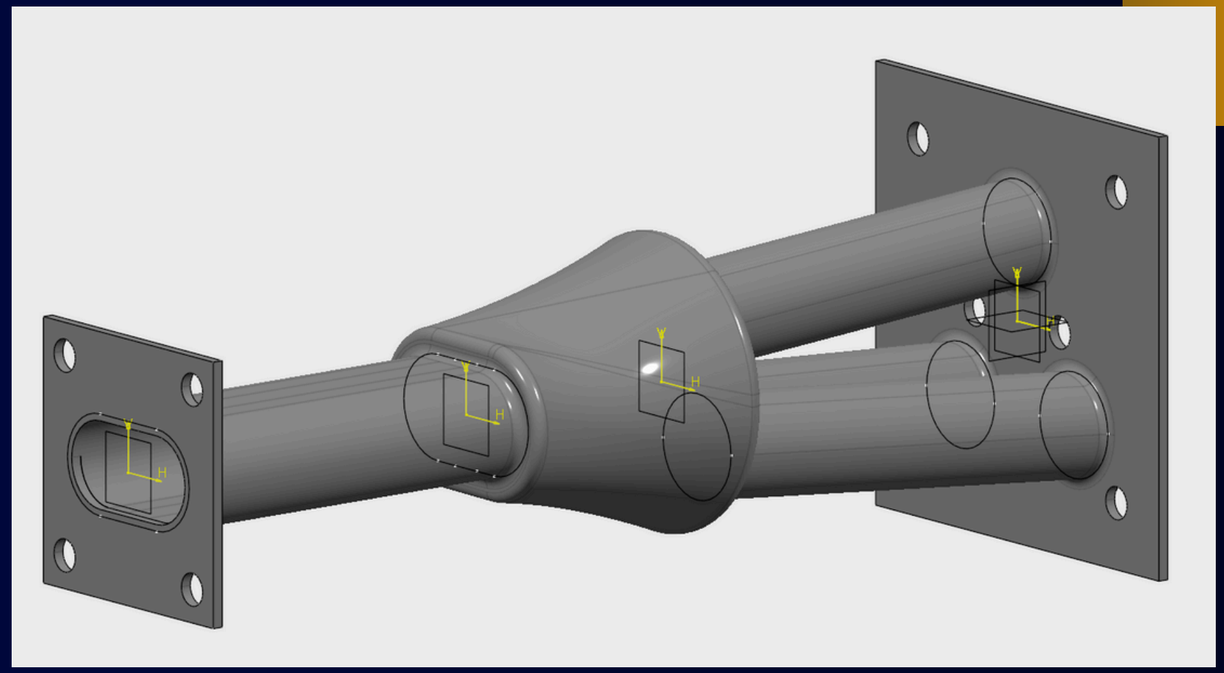
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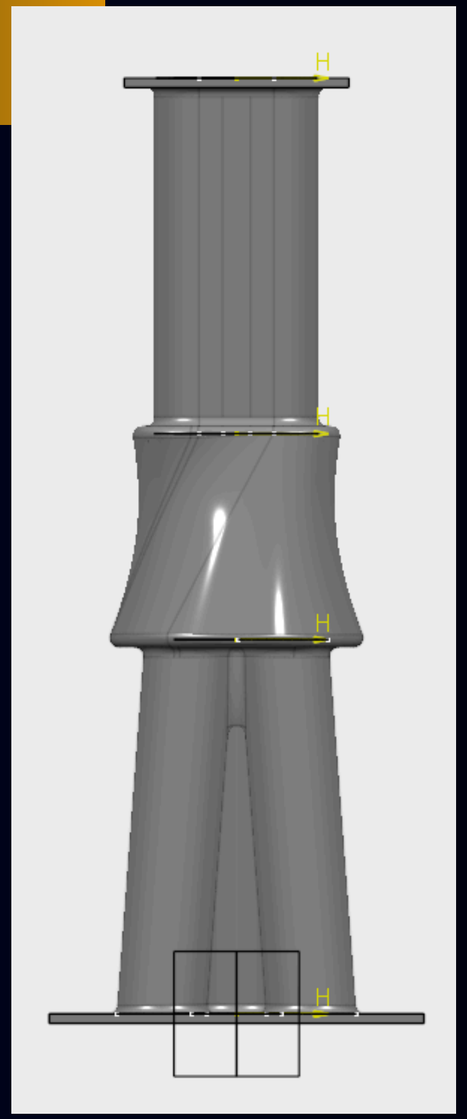
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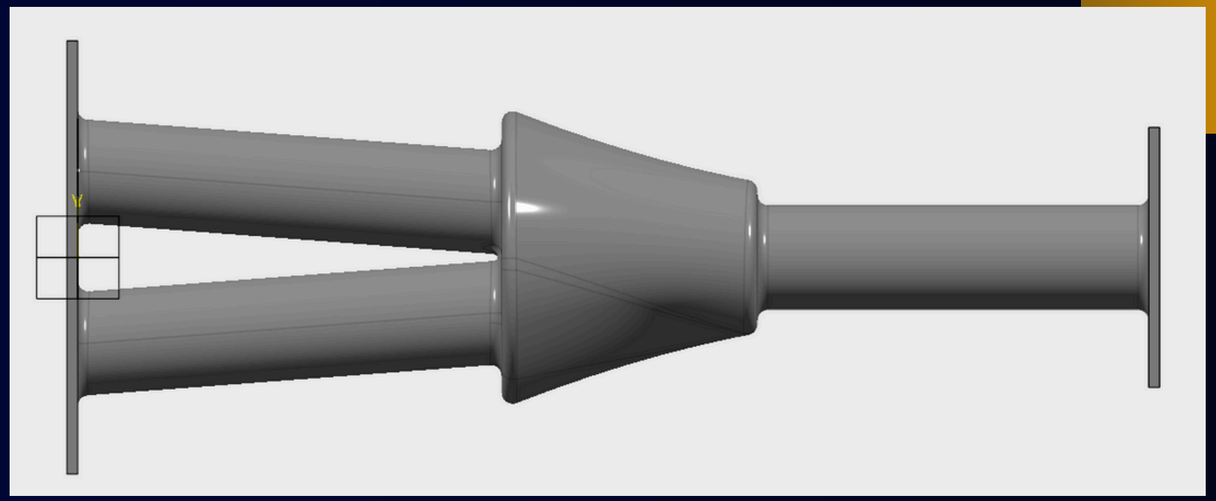


General conception

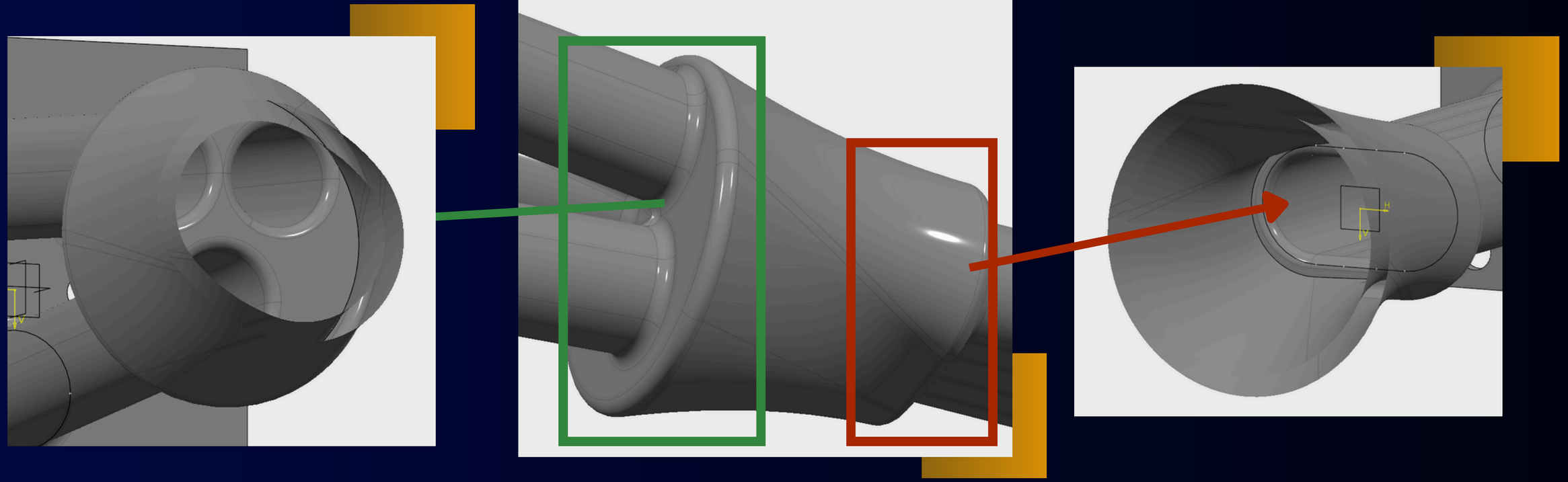
- Smooth connections at the inlets/outlets thanks to the “Multi-section solid” function
- Mixing chamber constructed in the same way, with volume adjustable at the click of a button (distance between the two planes)
- Screw fixings close to the inlets/outlets for a better seal
- 4 mm thick tubing, 8 mm thick mixing chamber







Side view



Cross-sectional views of the mixing chamber

Modal analysis

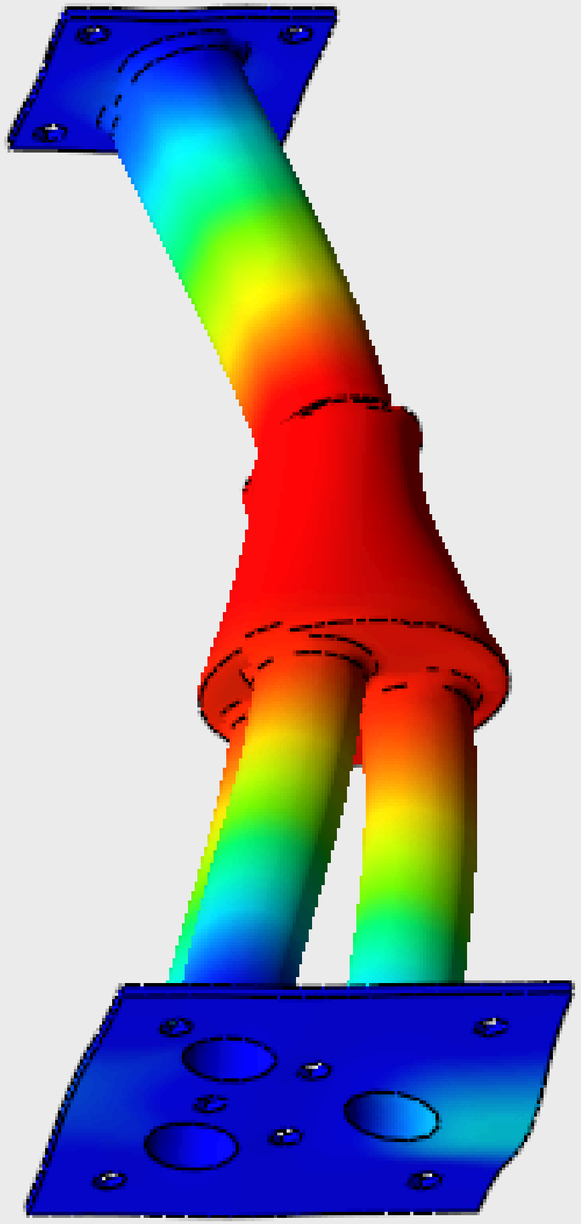
The natural frequency range of our hydraulic manifold complies with the specifications. The first natural frequency is relatively high, due to the rigidity of the steel used.

Eigenfrequencies (Hz) : 748,378 ; 906,105 ; 1486,864 ; 2153,396 ; 2614,689 ; ...

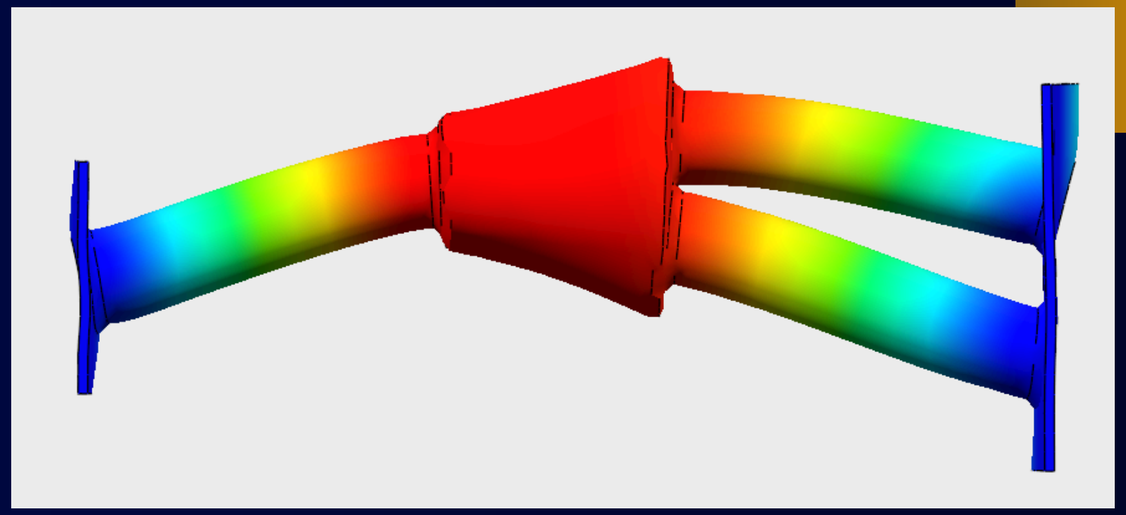
Further improvements

- Ribs along the pipes: increase the natural frequencies
- Securing the mixing chamber to a frame: eliminate any observed translational and torsional movements
- Add cross-bars between the inlet plate and the chamber: limit translational movement if securing to the frame is not possible

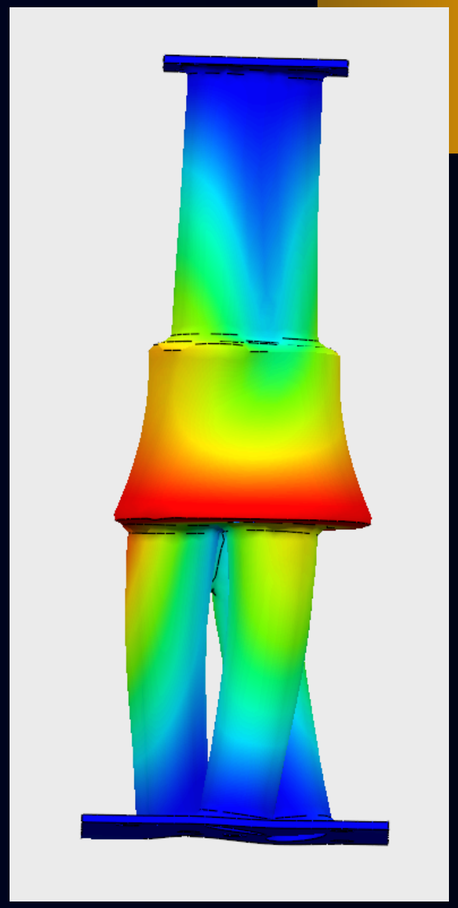
Representation of the first modal distortion



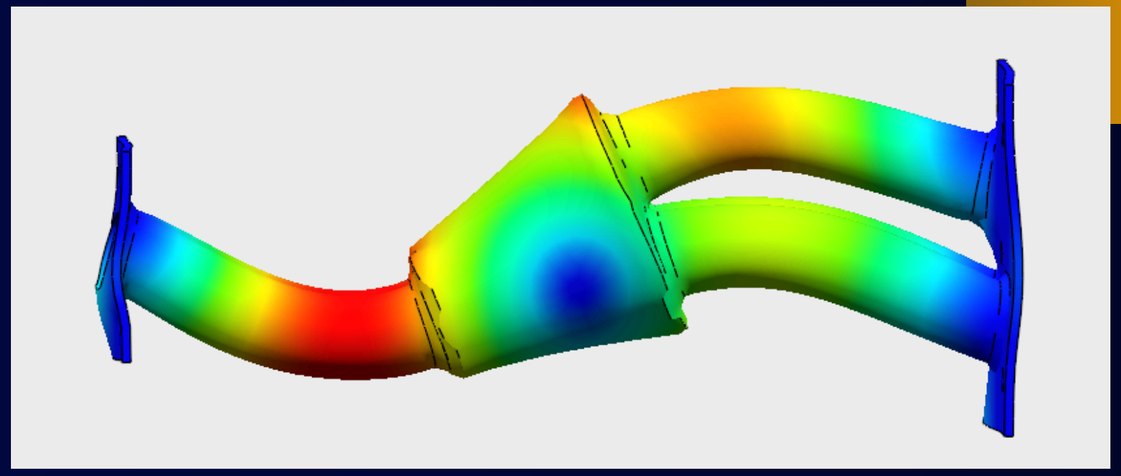
Mode 1: bending; the most dangerous for watertightness



Mode 3: torsion



Mode 4



Modal analysis: conclusions

Leak-tightness criterion:

- If the structure is subjected to loads at its natural frequencies, this may no longer be verified due to the displacement of the ends. There are several solutions:

    - Further shift the natural frequencies using the optimisation techniques described above (“To go further”)
    - Add fixings around sensitive inputs
    - Add gaskets

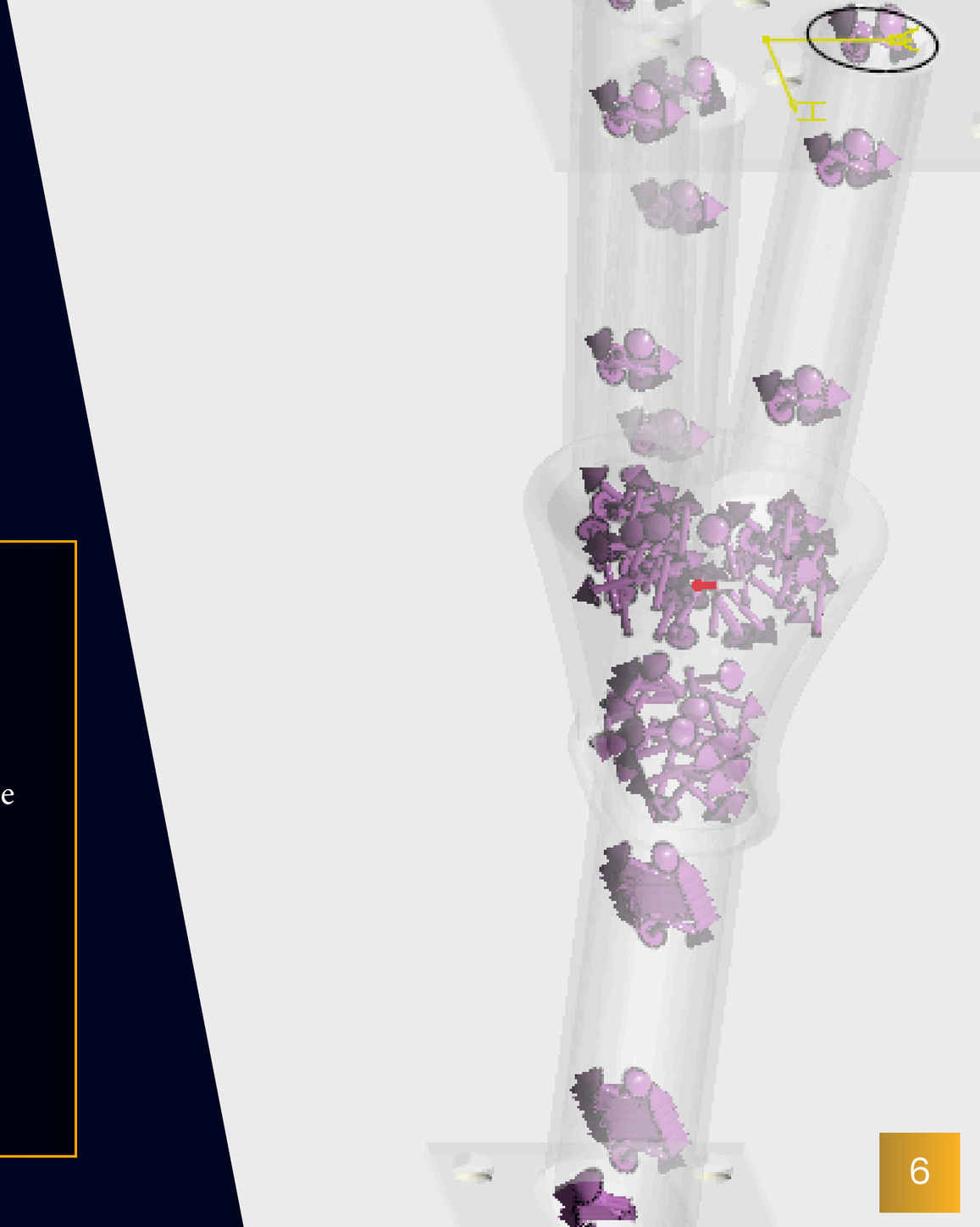
Pressure loading

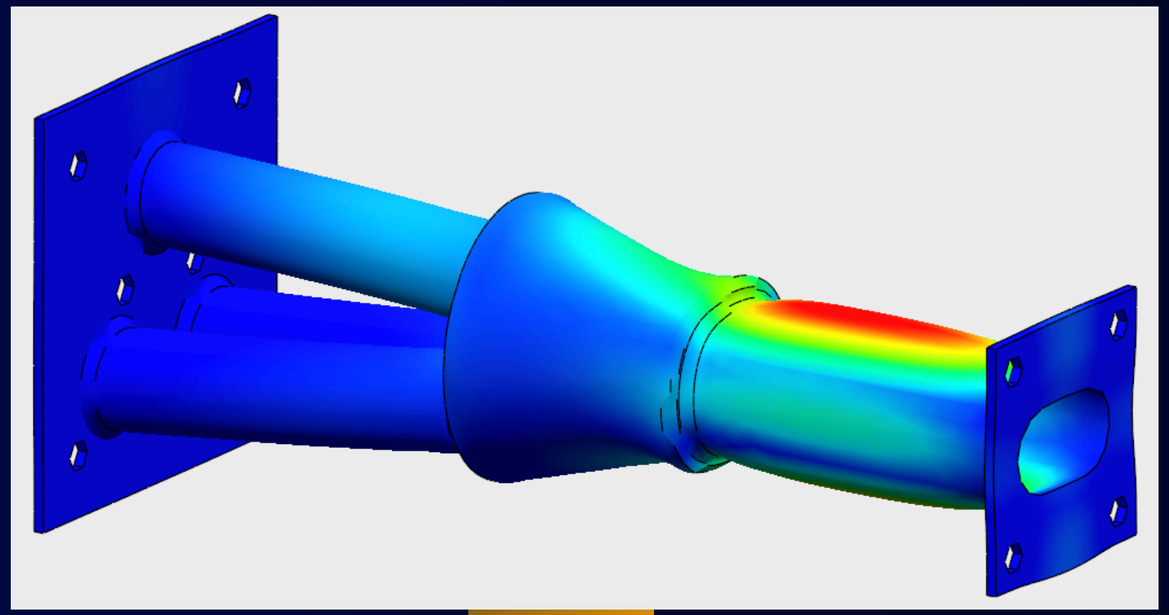
Boundary conditions: embedded in the threaded holes of the inlet/ outlet plates

As one might intuitively expect when pressurised, the pipes tend to swell

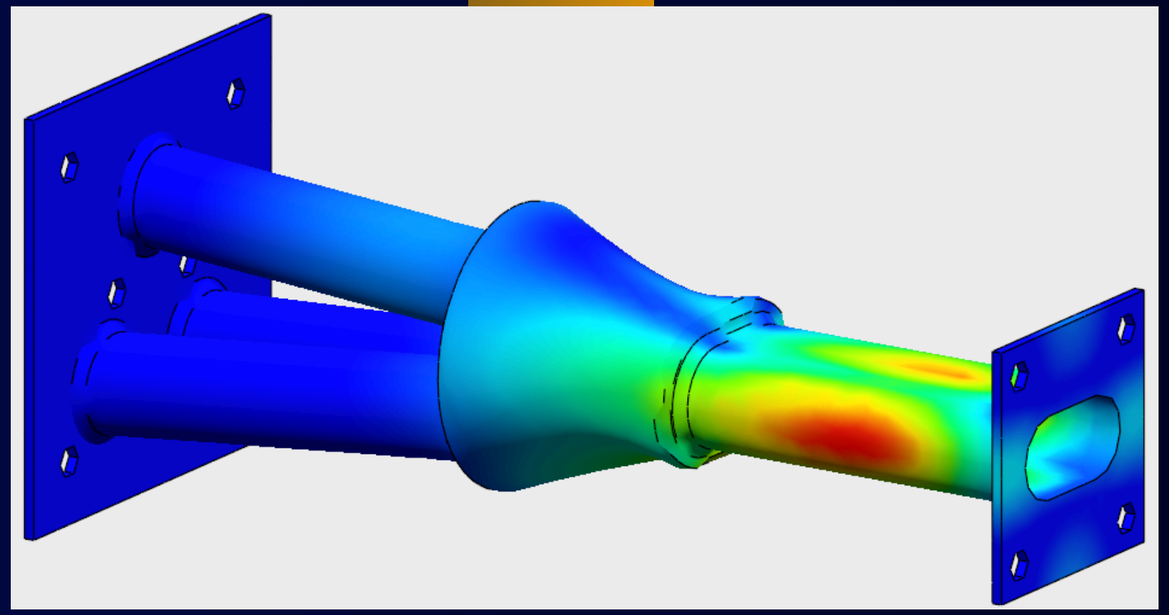
Optimisations carried out

- Straight pipes and corners rounded with fillets to mitigate stress concentration
- Reinforcement of the mixing chamber, which receives all the gases at once (!)
- Footprint of the mixing chamber = ¼ of the manifold's total footprint
- For a real-world manifold: a relatively long downstream pipe to allow the fluid to stabilise and return to laminar flow (~ Poiseuille)



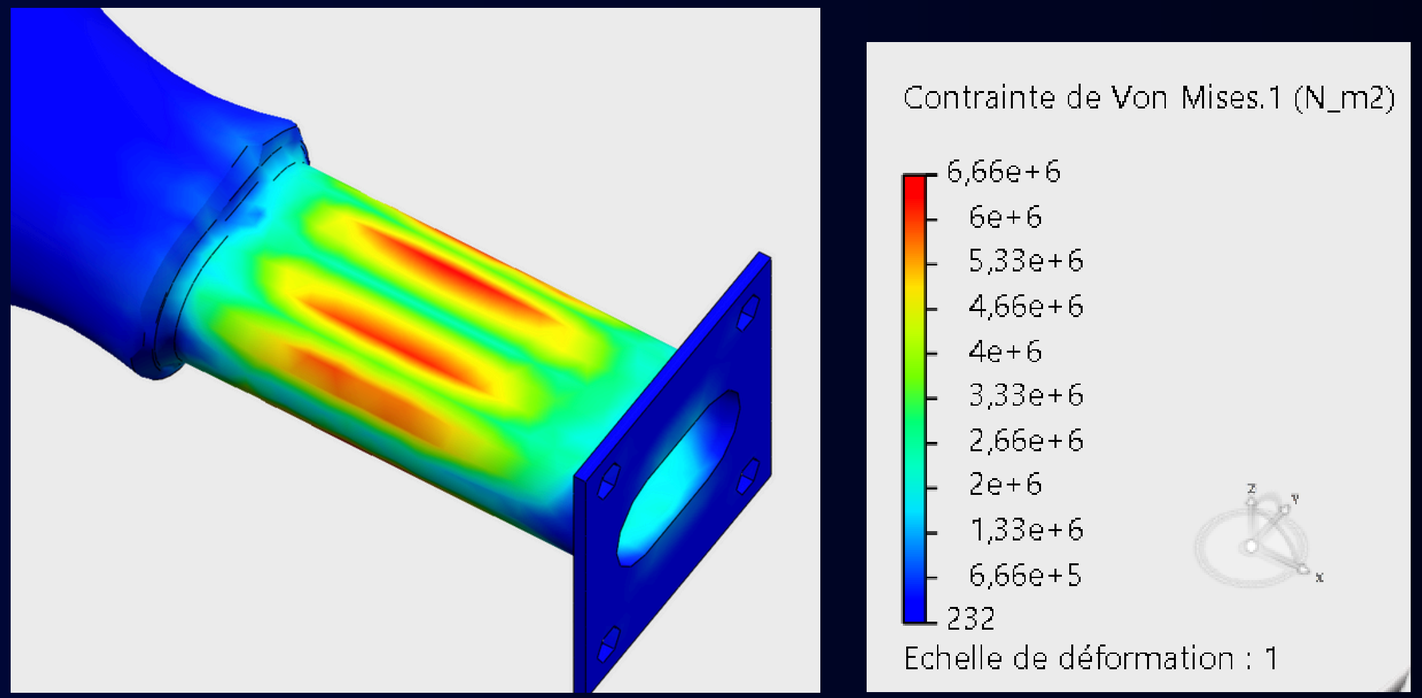


t=300s (maximum of cycle 3)

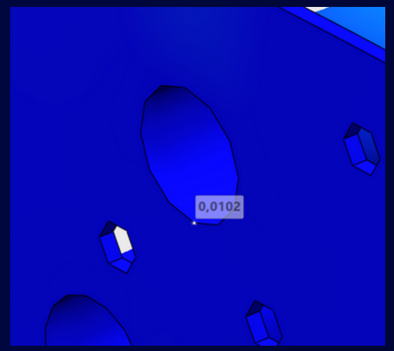


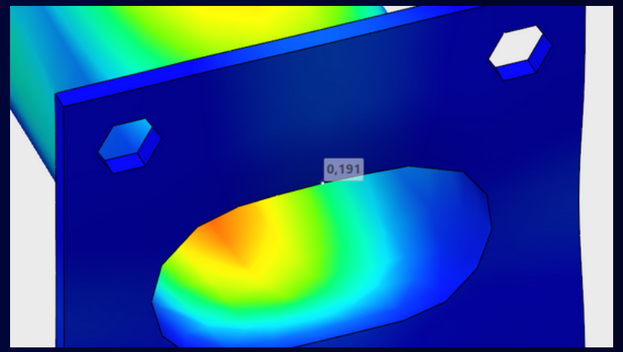
t=360s (end of cycle 3)

The swelling causes high stresses to build up in the downstream pipe. As the shape of the outlet does not allow the outlet flow rate to equal the inlet flow rate (hydraulic diameter ≠ 3 × inlet diameter), the fluid is expected to exert significant pressure on this section.



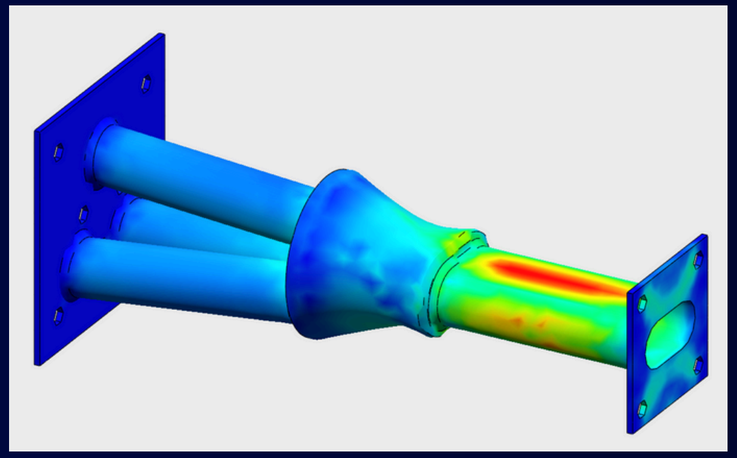
Compressive loading: conclusions

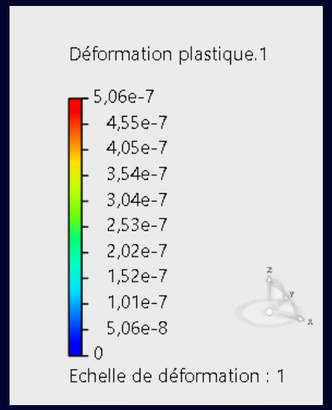




Sealing criterion: displacement of less than 0.5 mm

- To be on the safe side, screws and/or gaskets may be added





Plasticity: less than 0.3 per cent